

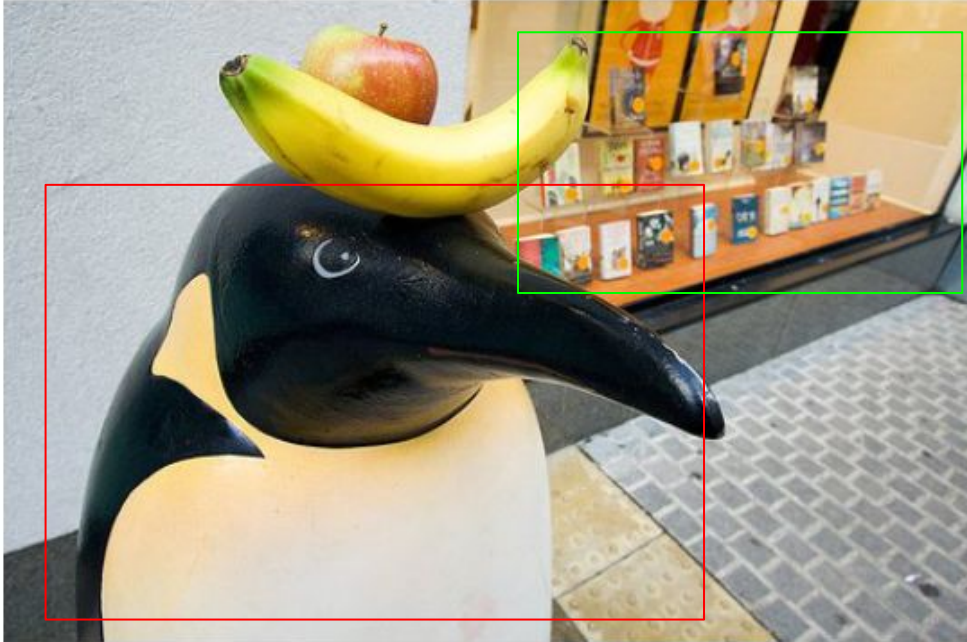
Disambiguation of One-Shot Visual Classification Tasks: A Simplex-Based Approach

Yassir BENDOU, Lucas DRUMETZ, Vincent GRIPON,
Giulia LIOI, Bastien PASDELOUP

`prenom.nom@imt-atlantique.fr`



The penguin, the banana and the library



Can we disentangle a one-shot task when multiple objects are present in a scene?

Introduction to Few-shot learning

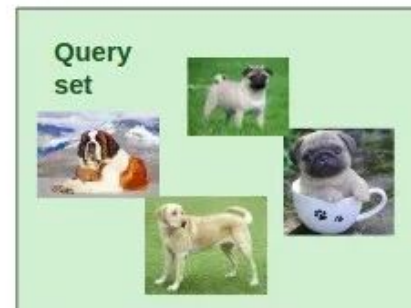
Some terminology...

In a Few-shot problem, we have:

- Nb of classes : **N-ways**.
- Support set : **K-shot** images par classe.

Typically:

5-ways | 1-shots or 5-shots

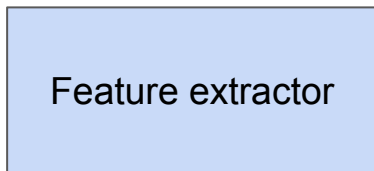


One few-shot problem

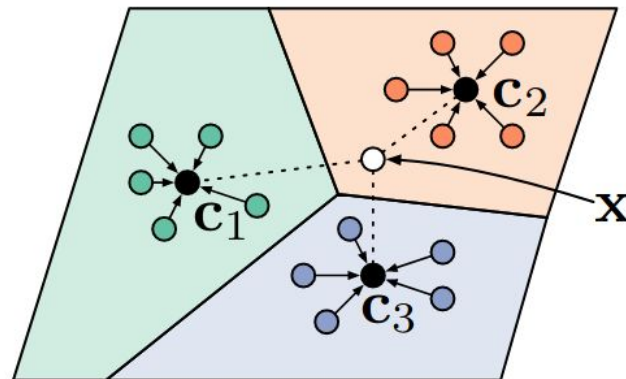
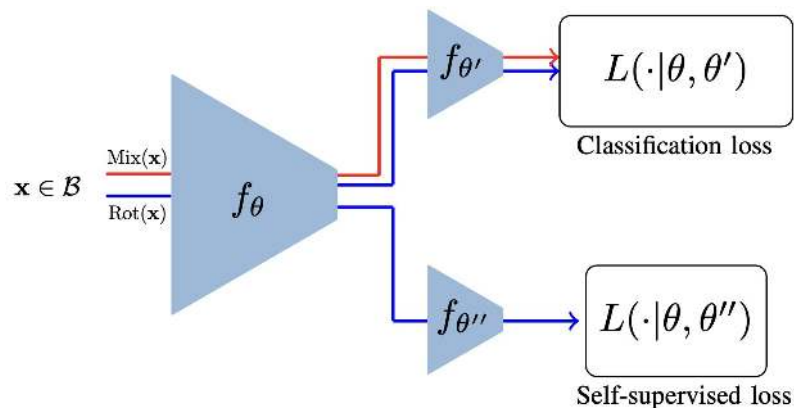
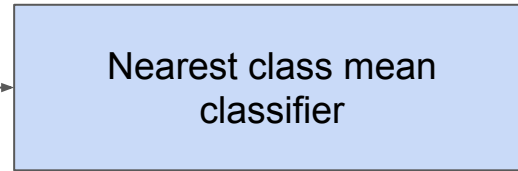
Introduction to Few-shot learning

In practice, we use a generic dataset

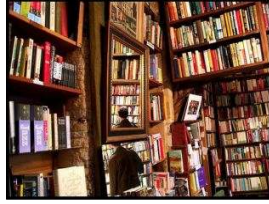
A generic dataset



A few-shot problem



The penguin, the banana and the library



Library



Armor



??



Hourglass



What was done so far?

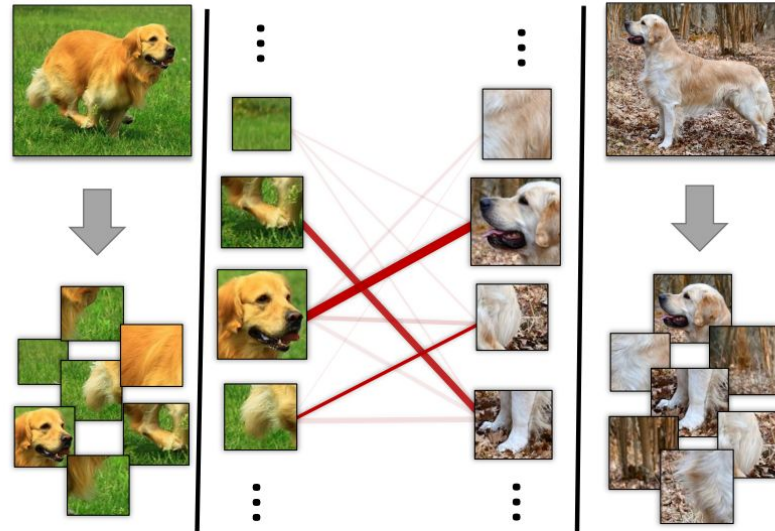
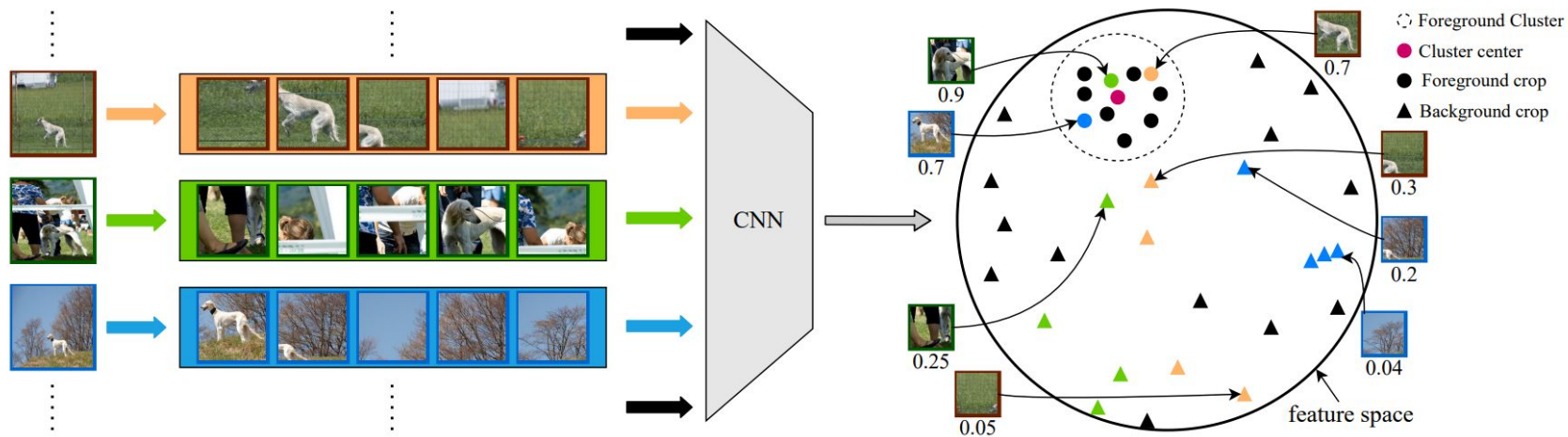


Figure 1: Illustration of using Earth Mover's Distance for one-shot image classification. Our algorithm uses optimal matching cost between image regions to represent the image distance.

DeepEMD:

https://openaccess.thecvf.com/content_CVPR_2020/papers/Zhang_DeepEMD_Few-Shot_Image_Classification_With_Differentiable_Earth_Movers_Distance_and_CVPR_2020_paper.pdf

What was done so far?



What was done so far?

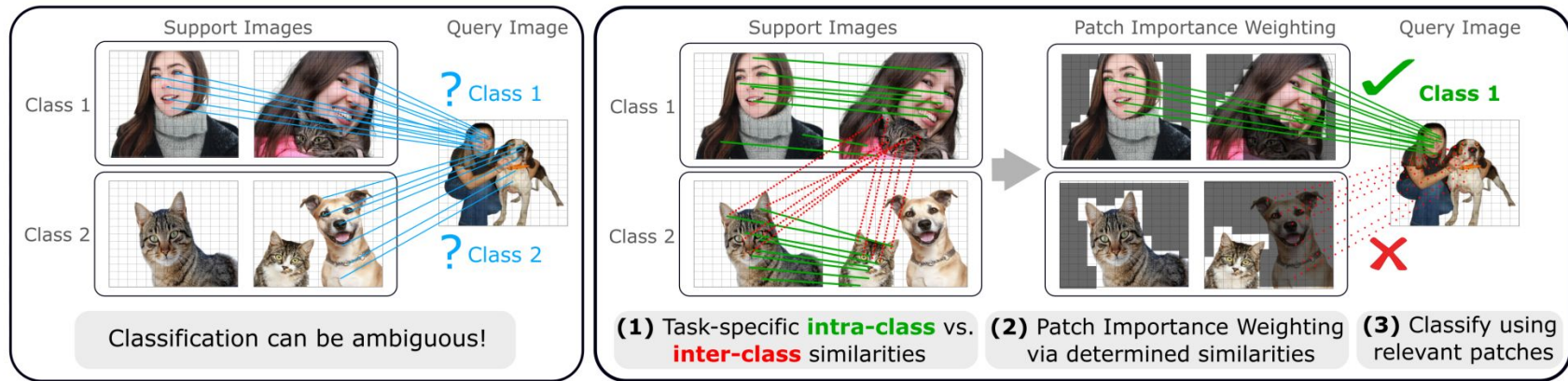
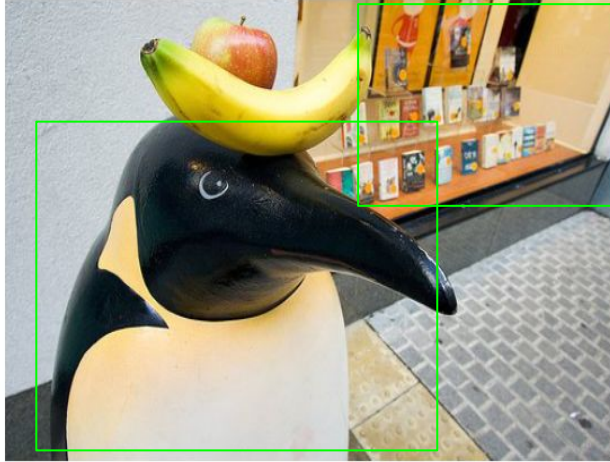
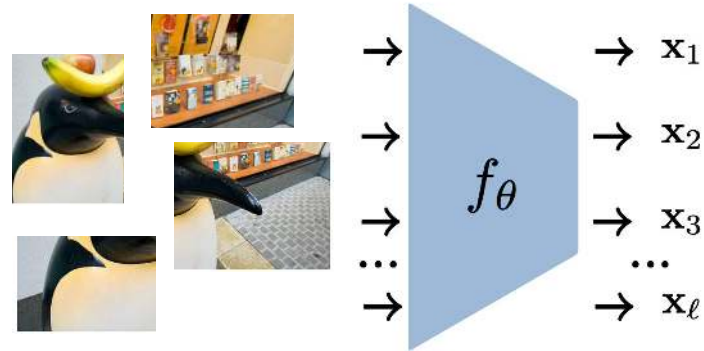


Figure 1: **Tackling classification ambiguity by interpreting images in context.** (Left): Labels assigned to real-world images with multiple entities only correctly describe a subset of the depicted content, leading to ambiguous classification results. (Right): Leveraging intra- and inter-class similarities and differences across the support set allows our method to determine the importance of each individual patch at inference time, i.e., to find out '*what matters most*' in each image. This information is then used to reweight support-query similarities and resolve ambiguity.

Our approach

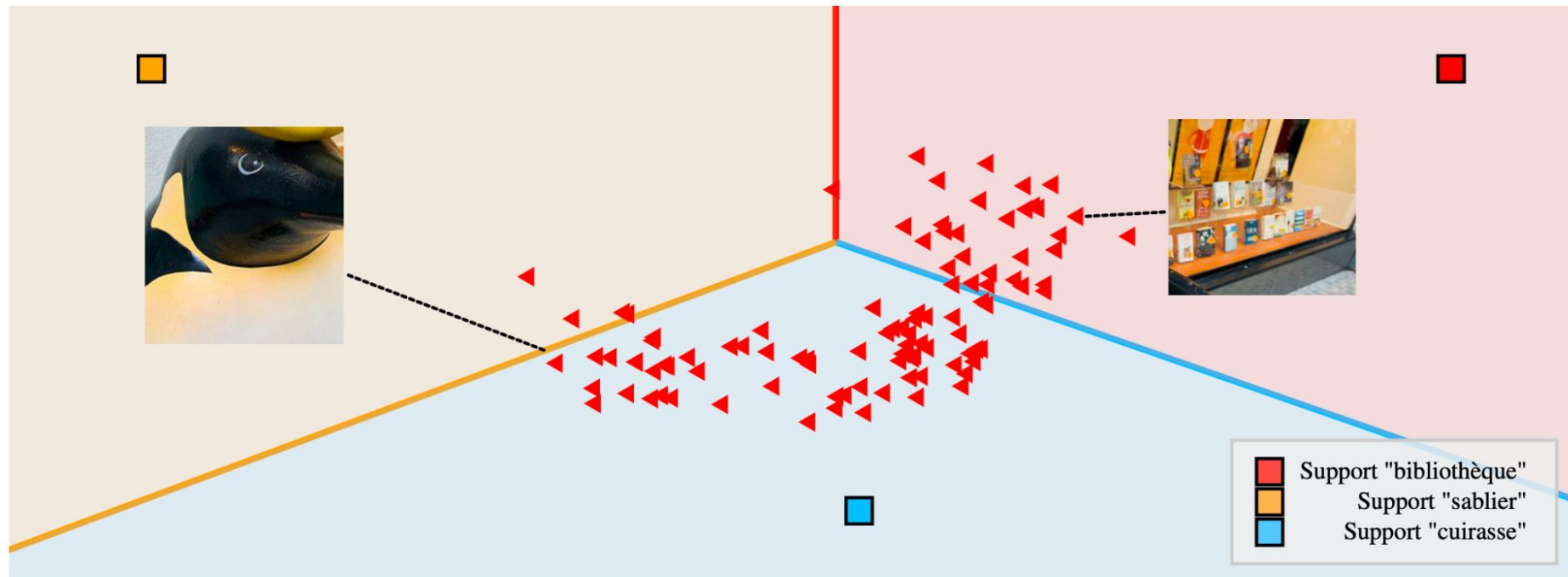


Different objects in an image have a different representation in the latent space.



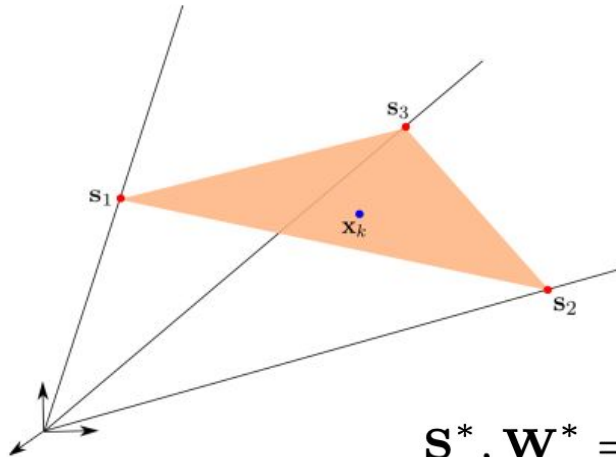
Feature extraction

Our approach: Visualizing the latent space



Projection of the latent space in a space spanned by the class centers “library”, “armor (cuirasse)” and “hourglass (sablier)”

Our approach



Simplex with 3 vertices

$$\mathbf{S}^*, \mathbf{W}^* =$$

$$\arg \min_{\mathbf{S}, \mathbf{W}} \quad \text{s.t.} \begin{cases} 0 \leq \mathbf{W} \\ \mathbf{W} \mathbf{1}_K = \mathbf{1}_N \end{cases}$$

$\mathbf{X}$: Matrix of features of multiple crops.

$\mathbf{S}$: Simplex with K vertices.

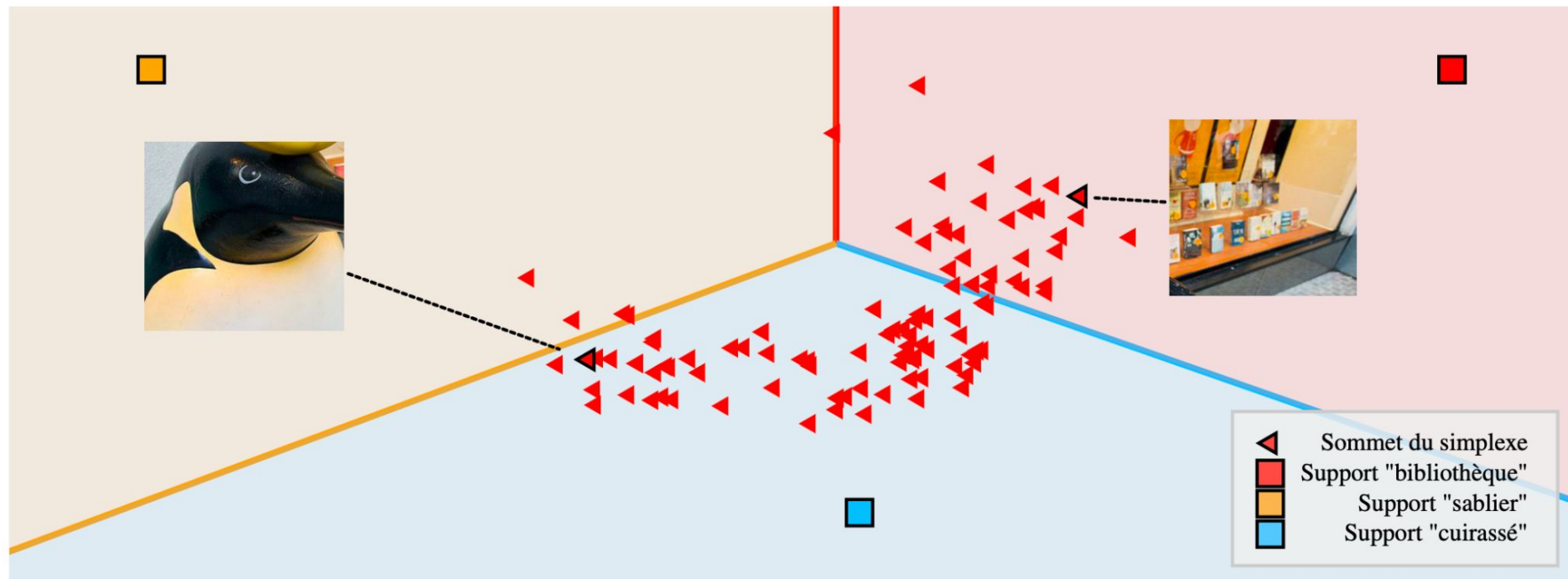
$\mathbf{W}$: Matrix of weights.

$$\left(\begin{array}{l} \lambda \sum_{k=1}^{K-1} \sum_{k'=k+1}^K \|\mathbf{S}_k - \mathbf{S}_{k'}\|_2^2 \\ + (1 - \lambda) \|\mathbf{W}\mathbf{S} - \mathbf{X}\|_2^2 \end{array} \right)$$

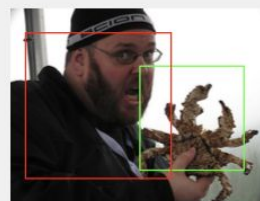
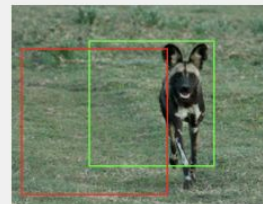
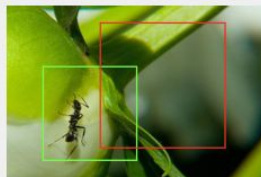
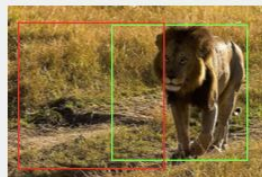
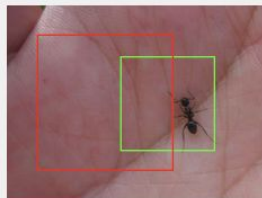
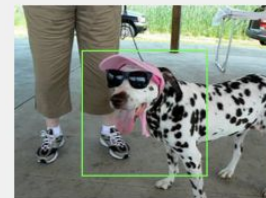
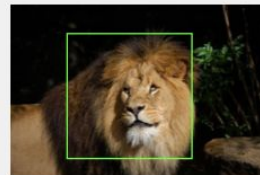
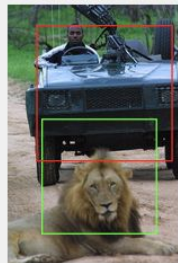
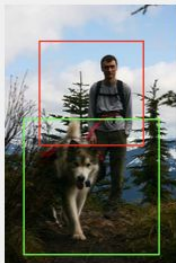
$$\hat{\mathbf{S}} = \left(\mathbf{W}^T \mathbf{W} + \mu \left(\mathbf{I}_K - \frac{\mathbf{1}\mathbf{1}^T}{K} \right) \right)^{-1} \mathbf{W}^T \mathbf{X}$$

$$\mu = N \frac{\lambda}{(K-1)(1-\lambda)}$$

Our approach: Visualizing the latent space



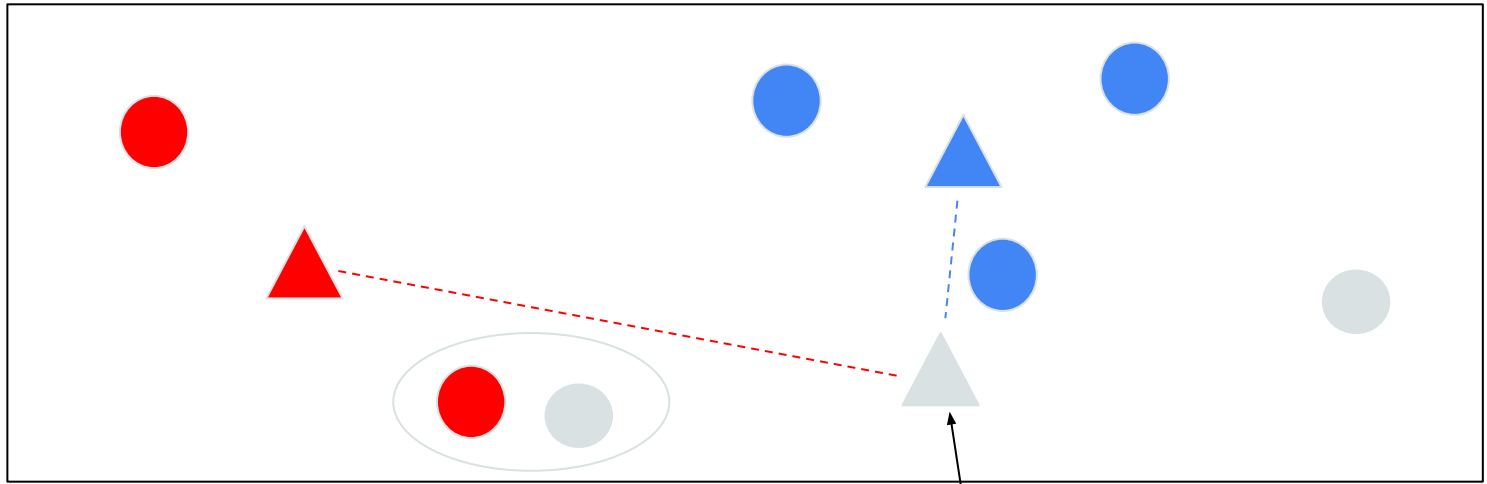
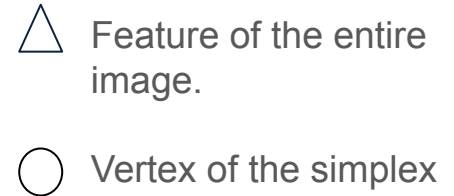
Visualisation des sommets extraits



Results of our method on minilmaNet

Classification

- First extract the vertices from each image.
- Classification : Identify the closest summit.



Query image for classification

Résultats

Mini-ImageNet 5-ways, 1-shots sur 100k Few-shot problems

Method	Performance (%)
Deep EMD v2 [6]	68.77 ± 0.29
PAL [12]	69.37 ± 0.64
invariance-equivariance [13]	67.28 ± 0.80
CSEI [14]	68.94 ± 0.28
COSOC [7]	69.28 ± 0.49
ASY [8]	70.77 ± 0.06
Notre méthode	70.90 ± 0.06

Table 1 - Mini-ImageNet one-shot performance

Limitations

- Crops are passed to the backbone individually, we don't take into account neighboring information for better feature representation.
- Problem not very occurring in minilImageNet
- The extracted vertices from the simplex were still close to each other in the latent space compared to other images.